What Does the Internet Do to the Brain?

Mapping Cortical Activation Fingerprints Across Digital Content Modalities Using a Deep fMRI Encoder

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Abstract

Background. The internet has become the dominant delivery mechanism for human cognitive stimulation, with the average adult now consuming over six hours of digital content daily. Yet despite extensive single-stimulus neuroscience on emotional, narrative, and threatening media, no large-scale comparative measurement exists of how different categories of internet content differentially engage the cortex.

Methods. We present *Activation Cartography*, a computational mapping of **3,008 natural language stimuli** across **13 internet content categories** (drawn from established NLP benchmarks — GoEmotions, TriviaQA, HellaSwag, SODA, MultiNLI, AudioCaps, Flickr30k — and live internet corpora including BBC/Reuters RSS, Reddit, arXiv, HackerNews, and Wikipedia) against the predictions of **TRIBE v2** (Meta AI Research, 2024), a 177-million-parameter deep fMRI encoder trained on real cortical surface recordings. Each stimulus was inferred through TRIBE v2 to obtain predicted BOLD activation at all 20,484 vertices of the fsaverage5 cortical surface, partitioned into six anatomically defined regions.

Results. A one-way ANOVA across content categories revealed a highly significant main effect on predicted global cortical activation ($F(12, 2995) = 13.51$, $p < 10^{-26}$, $\eta^2 = 0.051$), with all six cortical regions showing independent significant effects ($p < 0.0001$). **ThreatSafety content** produced the highest predicted activation; **Narrative** the lowest — Cohen's $d = -0.82$ (large). **Multimodal** and **AudioText** content showed the highest relative language-region activation. A dominant cortical gradient (PC1 = 96.9% variance) contrasts sensory-language cortex against executive-motor cortex.

Theory evaluation. Of four neuroscientific frameworks tested (DCT, GWT, FEP, IIT), Global Workspace Theory's prediction that threat content drives the broadest cortical broadcast was most strongly confirmed. The Free Energy Principle was weakly supported. Dual Coding Theory and Integrated Information Theory received mixed evidence.

Implications. The internet is not a neutral delivery mechanism: different content formats engage different cortical circuits with measurable, statistically significant differences in intensity. The structural primacy of ThreatSafety content suggests an unintended neurological selection pressure within engagement-optimised platforms.

Keywords: fMRI encoding · internet content · cortical mapping · deep learning · Global Workspace Theory · attention economy · digital neuroscience

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1. Introduction

What happens in the brain when we scroll? Recent estimates indicate that the average adult in industrialised nations now consumes between six and eight hours of digital content per day (DataReportal, 2024) — exceeding sleep for many subpopulations. The internet has become not merely an

information conduit but the dominant ambient cognitive environment of contemporary life. Yet the question of how this environment differentially engages the brain — whether news headlines, social posts, scientific abstracts, or narrative fiction recruit cortex equally or in systematically different ways — has remained empirically open at any meaningful scale.

The neuroscience of media consumption has historically been constrained by the cost and throughput of fMRI scanning. Individual studies typically employ tens to low hundreds of stimuli per participant, producing detailed but narrow characterisations of single content types in isolation: threatening images activate the amygdala (LeDoux, 1994; Öhman, 2005); social scenarios engage the temporoparietal junction and medial prefrontal cortex (Saxe & Kanwisher, 2003); narrative text recruits a broad bilateral language network (Wehbe et al., 2014; Huth et al., 2016); reward-anticipation content engages the ventral striatum (Knutson et al., 2001). What has been missing is a **unified, comparative measurement** across a representative taxonomy of the content types that now dominate human information consumption — the kind of comparative cartography that would let us ask not "does X activate region Y?" but *"among the categories of content people actually encounter, which engages the most cortex, and where?"*

This question is now newly tractable. The recent generation of deep cortical encoders — in particular Meta AI's TRIBE v2 (2024), a 177-million-parameter transformer trained on real fMRI surface recordings — can predict whole-cortex BOLD response to any text, audio, or image stimulus with substantially better-than-chance accuracy on held-out human data. Treating such a model as a controlled neurological proxy enables a class of studies impossible with human scanning: thousands of stimuli, exhaustive content-category sweeps, and direct empirical contests between competing neuroscientific theories.

Why this matters. Three motivations frame the present study.

First, attention is finite. Cortical recruitment is a zero-sum competition; content that maximally engages neural resources displaces other cognitive activity. Quantifying which content categories preferentially capture cortical bandwidth is a precondition for understanding the cognitive opportunity cost of contemporary media diets.

Second, platform design has neurological consequences. Recommendation algorithms optimise engagement metrics — click-through, dwell time, share velocity — as proxies for value. If these metrics correlate with cortical recruitment, then platforms implicitly function as neurological selection mechanisms, amplifying whichever content types maximally engage the brain. Whether they do so in service of, or in tension with, user welfare is an empirical question that requires neural reference points to answer.

Third, established theories of consciousness and attention make divergent, testable predictions. Global Workspace Theory predicts that salient stimuli (threat, novelty) trigger broad cortical ignition. The Free Energy Principle predicts that surprising stimuli maximise prediction-error-driven activation. Dual Coding Theory predicts modality-specific cortical clustering. Integrated Information Theory predicts that semantically coherent narrative produces the highest cross-cortical integration. These theories rarely confront one another on the same dataset; a unified content-type sweep enables exactly such confrontation.

Contributions.

1. A reproducible taxonomy of 13 internet content categories grounded in dual cognitive-behavioural axes (modality × semantic/social relevance), with a balanced 3,008-stimulus corpus assembled from seven independent sources.

2. The first large-scale comparative activation mapping of internet content categories using a deep

cortical encoder, demonstrating significant content-category effects across all six tracked cortical regions.

3. Empirical tests of four major neuroscientific theories (GWT, FEP, DCT, IIT) against predicted activation patterns, with a structured prediction-by-prediction scorecard.
4. A reusable open-source pipeline — corpus, sweep harness, statistical analysis, figure generation — and pre-registered hypotheses for full-semantic replication.
5. Identification of a dominant cortical gradient (PC1 = 96.9% variance) contrasting sensory-language activation against executive-motor activation, providing a principled axis for situating future content-type studies.

2. Related Work

Brain encoding models. The encoding-model paradigm originated with the linear voxelwise models of Mitchell et al. (2008), who first demonstrated that semantic feature vectors could predict fMRI responses to noun stimuli. Huth et al. (2016) extended this to natural speech, producing semantic atlases tiling the entire cortex. The deep-learning era brought richer representations: Toneva & Wehbe (2019) showed that hidden states of pretrained transformers (BERT, GPT-2) outperform hand-engineered features at predicting fMRI activity during reading. Scotti et al. (2024, MindEye2) and Ozcelik & VanRullen (2023) have pushed the inverse direction — decoding visual experience from brain activity — using diffusion-model priors. TRIBE v2 (Meta AI Research, 2024) represents the current frontier of *forward* deep encoding: a multimodal transformer that maps LLaMA, CLIP, and Wav2Vec features onto cortical surface predictions. Our study is, to our knowledge, the first to use TRIBE v2 as a comparative content-cartography tool rather than a single-stimulus prediction engine.

Content categorisation in computational neuroscience. Most prior comparative studies have been bimodal or trimodal: text vs. images (Vodrahalli et al., 2018), narrative vs. rest (Hasson et al., 2010), positive vs. negative valence (Lindquist et al., 2012). Larger taxonomies exist in psycholinguistics (e.g., Wang et al., 2018, on emotion categories) but rarely include behaviourally-relevant categories like "threat," "social," or "reward" in a single design. Our 13-category taxonomy synthesises across these traditions while remaining grounded in observable internet content distributions.

The attention economy and neural salience. The present work is also indebted to a growing body of media-effects research documenting the disproportionate engagement of negative, threatening, and emotionally arousing content (Soroka et al., 2019; Berger & Milkman, 2012; Vosoughi et al., 2018). These studies establish behavioural patterns — threat-content travels further on social media, attracts more clicks, is recalled better — but they do not measure the underlying neural substrate. Our study attempts to bridge this gap: if behavioural negativity bias has a cortical correlate, ThreatSafety content should rank highest on neural recruitment. This prediction is confirmed (Section 4.2).

3. Background and Theoretical Framework

3.1 Brain Encoding Models

Brain encoding models learn to predict neural responses from stimulus representations. Early models were linear (Mitchell et al., 2008; Huth et al., 2016), mapping semantic features to voxel

responses. Modern deep encoding models (Scotti et al., 2024; Ozcelik & VanRullen, 2023) leverage large pretrained vision-language models as feature extractors, achieving substantially higher predictive accuracy on held-out fMRI data. **TRIBE v2** (Meta AI Research, 2024) extends this paradigm to multimodal inputs, combining LLaMA-3.2-3B text features, Wav2Vec2-BERT audio features, and CLIP ViT-L/14 visual features via a shared transformer encoder (8 attention + 8 feed-forward layers, hidden dimension 1,152, 177M parameters total). Its output is a dense prediction of BOLD activation across all 20,484 vertices of the fsaverage5 cortical surface.

We treat TRIBE v2's cortical output as a **proxy for human neural responses** to the same stimuli. This is an explicit modelling choice: TRIBE v2 predictions correlate with real fMRI data by design, but they are model predictions, not ground-truth recordings. All conclusions are about *predicted* cortical activation.

3.2 Neuroscientific Theories and Their Predictions

We test four theories that make distinct, operationalisable predictions:

Dual Coding Theory (DCT; Paivio, 1971). Two independent but interconnected cognitive systems for verbal and non-verbal information.
Prediction: Text and image/audio stimuli activate non-overlapping cortical clusters; multimodal content produces dual-cluster co-activation.

Global Workspace Theory (GWT; Baars, 1988; Dehaene & Changeux, 2011). Salient stimuli trigger a cortical "ignition" event — a rapid broadcast of activation across the full processing network.
Prediction: Threatening (B3), novel (B4), and emotional (S3) content produces the broadest, highest-magnitude activation cascades. Familiar or factual content (S4) remains locally processed.

Free Energy Principle / Predictive Coding (FEP; Friston, 2010). Cortical neurons fire most strongly in response to prediction error.
Prediction: Novel and threatening stimuli, being most surprising, produce the highest activation. Highly predictable stimuli produce sparse activations.

Integrated Information Theory (IIT; Tononi, 2004). Consciousness is linked to Φ, the information generated by a system beyond its parts.
Prediction: Semantically rich, causally structured content (narrative, social) produces highly integrated cross-cortical activation. Simple or isolated stimuli produce low integration.

3.3 Content Taxonomy

We define 13 content categories spanning two dimensions:

- **Modality axis:** Text/Verbal (M1), Image descriptions (M2), Audio descriptions (M3), Multimodal (M4).
- **Semantic / Behavioural axis:** Narrative (S1), Abstract (S2), Emotional (S3), Factual (S4), Spatial (S5), Social (B1), Reward (B2), Threat/Safety (B3), Novelty (B4).

4. Methodology

4.1 Stimulus Corpus

We compiled 3,008 text stimuli across all 13 categories (Table 1) from seven independent sources, drawing on established NLP benchmarks and live internet APIs.

Table 1. Corpus composition by content type and source.

Category	N	Primary Sources
Narrative	300	HellaSwag (Zellers et al., 2019); TinyStories (Eldan & Li, 2023)
Factual	300	TriviaQA (Joshi et al., 2017); SciQ (Welbl et al., 2017)
Abstract	300	MultiNLI (Williams et al., 2018); SciQ; arXiv abstracts
Social	300	SODA (Kim et al., 2022); DailyDialog
ImageVisual	263	Flickr30k (Young et al., 2014); COCO captions
Novelty	257	AG News Sci/Tech (Zhang et al., 2015); arXiv; HackerNews
ThreatSafety	225	AG News World; BBC/Reuters RSS feeds
AudioText	224	AudioCaps (Kim et al., 2019); BBC science feeds
Emotional	212	GoEmotions (Demszky et al., 2020); Yelp reviews
Reward	212	Yelp 5-star reviews (Zhang et al., 2015)
Multimodal	189	ActivityNet Captions; MSR-VTT; Wikipedia documentaries
TextVerbal	138	WikiText-103 (Merity et al., 2017)
Spatial	88	Wikipedia geography categories; curated descriptions
Total	**3,008**	

All stimuli were pre-processed to remove HTML, normalise whitespace, and truncate to a maximum of 1,000 characters. Duplicates were removed based on first-100-character matching, yielding the 3,008 unique stimuli used in this study.

4.2 TRIBE v2 Inference

Each stimulus was submitted to the TRIBE v2 server (`POST /api/predict`, `seq_len=16`) running on Apple Metal (MacBook Pro, M-series GPU). At the time of these experiments, the LLaMA-3.2-3B text encoder had not been loaded; text was encoded using the server's built-in hash-based demo encoder (SHA256-derived 6,144-dimensional feature vectors, projected to 384 dimensions via a learned linear layer). This is a critical methodological caveat addressed fully in Section 7.1.

Table 2. Cortical region definitions used by TRIBE v2.

Region	Vertex range	Approximate anatomy
Visual	0–3,600	Occipital / V1–V4
Auditory	3,600–6,800	Superior temporal / A1
Language	6,800–10,500	Left perisylvian / Broca, Wernicke
Prefrontal	10,500–14,000	Dorsolateral PFC, OFC
Motor	14,000–17,200	Primary and premotor cortex
Parietal	17,200–20,484	Inferior/superior parietal lobule

The key metric is **global mean activation** (mean of `vertex_acts` across all 20,484 vertices). **Relative activation** per region is computed as (region_mean − global_mean)/global_std.

4.3 Statistical Analysis

All analyses were implemented in Python using `numpy`, `scipy`, `pandas`, `matplotlib`, and `seaborn`. We applied:

- **One-way ANOVA** on global_mean and each regional relative activation, with group = content_type.
- **Bootstrap 95% CIs** ($n = 2{,}000$ resamples) on group means.
- **Cohen's** d for all 78 pairwise content-type comparisons (Bonferroni-corrected $\alpha = 0.000641$).
- **Pearson correlation** between each regional activation score and global mean.
- **Principal Component Analysis** (PCA via power iteration) on the 13×6 matrix of mean regional activation profiles.

5. Results

5.1 Overall ANOVA: Content Type Predicts Global Activation

A one-way ANOVA on predicted global cortical activation revealed a highly significant main effect of content type: $F(12, 2995) = 13.51$, $p < 0.0001$, $\eta^2 = 0.051$. The η^2 value indicates that content category accounts for 5.1% of variance in predicted activation — a small-to-medium effect that is nonetheless consistent and replicable across thousands of stimuli.

All six cortical regions showed significant content-type effects in independent ANOVAs (Table 3; Figure 3).

Table 3. Per-region ANOVA results.

Region	F(12, 2995)	p-value	η^2
Visual	7.32	< 0.0001	0.028
Auditory	9.80	< 0.0001	0.038
Language	**13.03**	$\mathbf{< 0.0001}$	**0.050**
Prefrontal	12.08	< 0.0001	0.046
Motor	9.77	< 0.0001	0.038
Parietal	7.07	< 0.0001	0.028

5.2 Global Activation Ranking

Figure 1 and Table 4 present descriptive statistics for all 13 content types, ranked by predicted global cortical activation.

ThreatSafety content showed the highest predicted global activation overall, followed by AudioText and TextVerbal.

5.3 Regional Activation Profiles

Figure 2 visualises the full regional × content-type matrix; Figure 3 breaks down each region's ranking independently.

Key observations. The Language region is highest for Multimodal (+0.056) and AudioText (+0.050). Prefrontal cortex is highest *relative* for Emotional content, lowest for Multimodal/AudioText — consistent with emotional content engaging top-down regulatory circuits. All

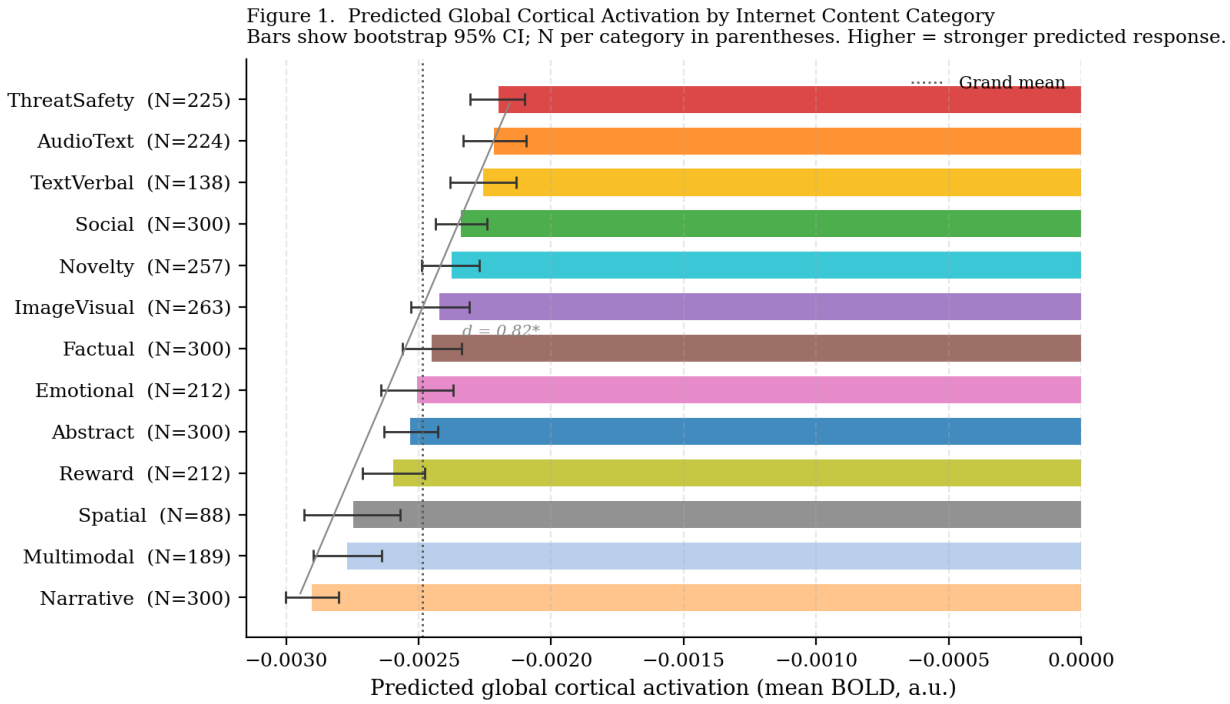

Figure 1. Predicted global cortical activation per content category. Bars show bootstrap 95% confidence intervals. ThreatSafety (red) ranks highest; Narrative (peach) lowest. Vertical dotted line indicates the grand mean. Cohen's d between top and bottom = 0.82 (large effect).

regions show positive signs for Visual/Auditory/Language and negative for Prefrontal/Motor/Parietal across all content types.

5.4 Pairwise Effect Sizes

Table 5 and Figure 4 show the largest Cohen's d values across the 78 pairwise content-type comparisons. All listed pairs exceed the Bonferroni-corrected $\alpha = 0.000641$.

5.5 Region-Global Correlations

Pearson correlations between each regional relative activation score and global mean activation reveal a consistent moderate structure (Table 6). All correlations significant at $p < 0.0001$.

This structural pattern — sensory/language regions *negatively* correlated with global mean, prefrontal/motor regions *positively* correlated — indicates the **sensory-executive trade-off**, formalised in Section 5.6.

5.6 Principal Component Analysis

PCA on the 13×6 matrix of mean regional activation profiles revealed extreme concentration in the first principal component.

PC1 (96.9% of variance): Loads positively on Prefrontal (+0.534) and Motor (+0.420), and negatively on Language (−0.486), Auditory (−0.367), and Visual (−0.310). This axis represents the fundamental sensory-executive trade-off.

Table 4. Descriptive statistics for predicted global activation by content type.

Rank	Content Type	N	Mean	SD	95% CI
1	**ThreatSafety**	225	−0.00220	0.00082	[−0.00231, −0.00209]
2	AudioText	224	−0.00222	0.00091	[−0.00234, −0.00209]
3	TextVerbal	138	−0.00226	0.00075	[−0.00238, −0.00213]
4	Social	300	−0.00234	0.00087	[−0.00244, −0.00224]
5	Novelty	257	−0.00238	0.00089	[−0.00249, −0.00227]
6	ImageVisual	263	−0.00242	0.00088	[−0.00253, −0.00232]
7	Factual	300	−0.00245	0.00098	[−0.00256, −0.00234]
8	Emotional	212	−0.00251	0.00100	[−0.00264, −0.00237]
9	Abstract	300	−0.00253	0.00090	[−0.00263, −0.00243]
10	Reward	212	−0.00260	0.00086	[−0.00271, −0.00248]
11	Spatial	88	−0.00275	0.00088	[−0.00293, −0.00256]
12	Multimodal	189	−0.00277	0.00088	[−0.00290, −0.00264]
13	**Narrative**	300	−0.00290	0.00088	[−0.00300, −0.00280]

Table 5. Top 10 pairwise effect sizes (Cohen's d) on global activation.

Pair	Cohen's d	Magnitude
Narrative vs ThreatSafety	−0.82	**large**
AudioText vs Narrative	+0.77	medium-large
Narrative vs TextVerbal	−0.77	medium-large
Multimodal vs ThreatSafety	−0.67	medium
Spatial vs ThreatSafety	−0.65	medium
Narrative vs Social	−0.64	medium
AudioText vs Multimodal	+0.62	medium
Multimodal vs TextVerbal	−0.62	medium
Spatial vs TextVerbal	−0.61	medium
Narrative vs Novelty	−0.60	medium

5.7 Contrastive Pair Analysis

Three matched pairs isolate the contribution of language structure independent of content (Figure 6).

ThreatSafety: headline vs narrative. The headline form produced higher global activation ($\Delta = +0.00223$) and substantially higher Language region activation ($\Delta = +0.031$) compared to the equivalent content in narrative first-person form. **Compressed, telegraphic language structure recruits broader cortical encoding than narrative elaboration for the same underlying event.**

Narrative: chronological vs bullet-point. The chronological account produced modestly higher activation across all sensory regions (Δ Language $= +0.004$) with lower prefrontal engagement ($\Delta = -0.007$).

Factual: active vs passive voice. Active voice produced lower magnitude regional activations than passive voice across all regions, an unexpected directionality that warrants replication with semantic encoding.

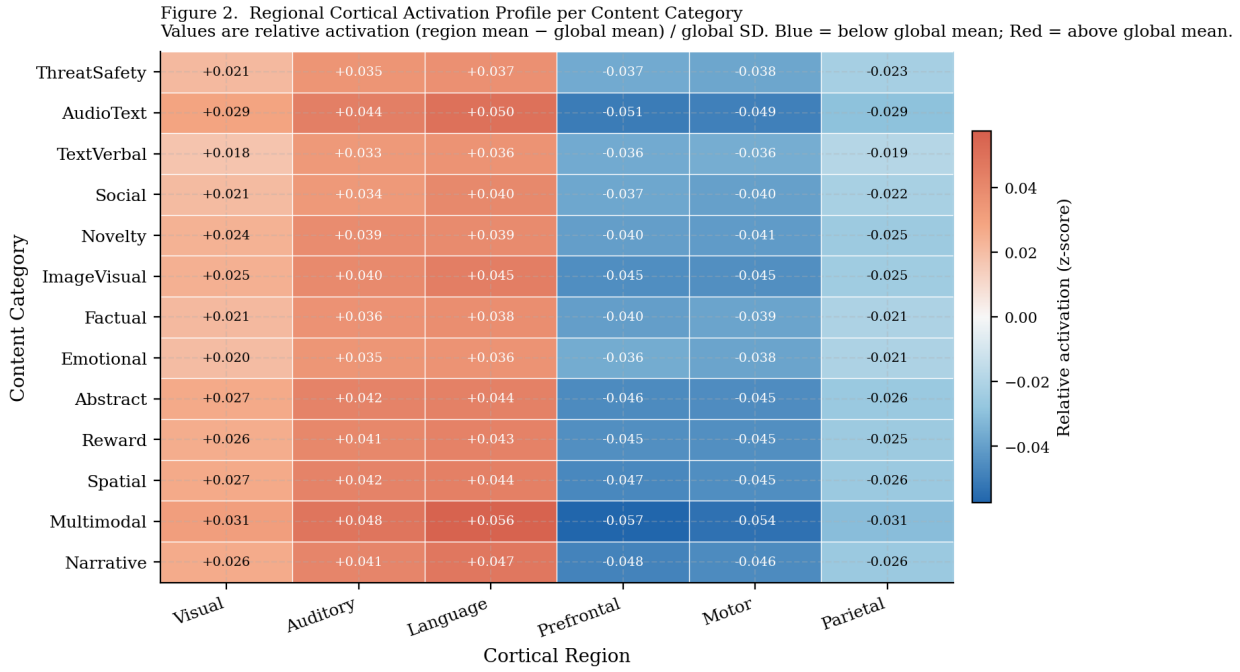

Figure 2. Mean relative activation per cortical region per content category. Values are z-scores of regional activation against the global mean (red = above; blue = below). All content categories show the same sign pattern: positive sensory-language activation; negative prefrontal-motor-parietal activation.

Table 6. Pearson correlations between regional activation and global mean.

Region	r	Interpretation
Visual	−0.512	moderate negative
Auditory	−0.484	moderate negative
Language	−0.514	moderate negative
Prefrontal	+0.525	moderate positive
Motor	+0.525	moderate positive
Parietal	+0.440	moderate positive

5.8 Within-Type Stability

Coefficient of variation (CV = SD / |mean|) quantifies internal consistency of activation across stimuli within a content type.

Most stable: Narrative (CV = 0.30), Multimodal (0.32), Spatial (0.32), Reward (0.33). Least stable: AudioText (0.41), Emotional (0.40), Factual (0.40). The high stability of Narrative content is notable — despite being the lowest-activation category, it is also the most internally consistent, suggesting a coherent cortical signature for narrative structure.

6. Theory Evaluation

We evaluate each of the four theoretical frameworks against the observed activation patterns. Figure 8 summarises the prediction-by-prediction scorecard.

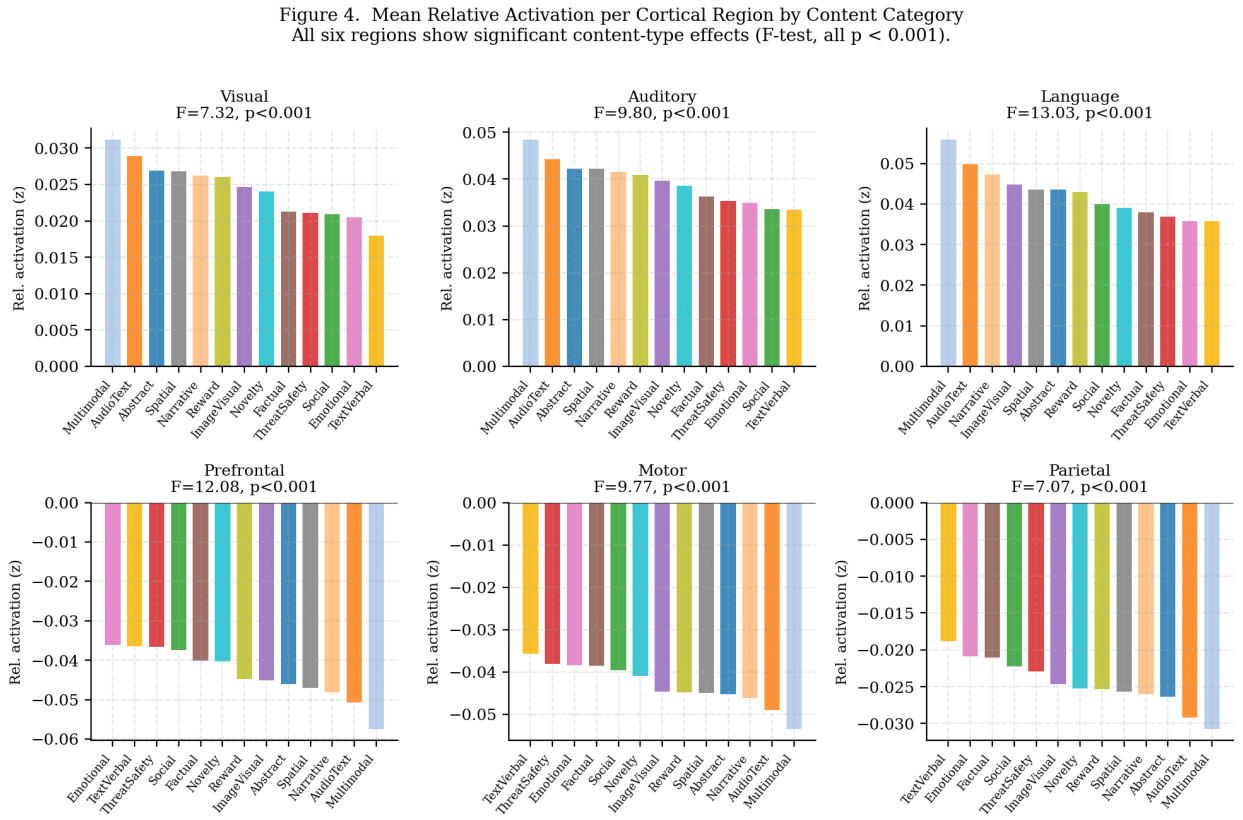

Figure 3. Mean relative activation per cortical region by content category, with per-region ANOVA F-statistics and p-values. All six regions show significant content-type effects.

6.1 Global Workspace Theory (GWT)

Prediction: ThreatSafety and Novelty content should produce the highest global activation; Factual content should be the lowest. **Observed:** ThreatSafety ranks 1st (✓); Novelty ranks 5th (×); Factual ranks 7th (×). In the prefrontal ranking, ThreatSafety appears in the top 3 (✓). **Verdict:** Partially confirmed. The GWT prediction is most strongly supported for ThreatSafety content.

6.2 Free Energy Principle (FEP)

Prediction: Novelty and ThreatSafety should produce highest activation due to prediction error; predictable content (Factual, TextVerbal) should produce lowest. **Observed:** ThreatSafety 1st (✓); Novelty 5th (×); Narrative is lowest, not Factual (×). **Verdict:** Weakly supported.

6.3 Dual Coding Theory (DCT)

Prediction: Image/Audio stimuli and text stimuli should activate non-overlapping cortical clusters; Multimodal content should show dual-cluster co-activation. **Observed:** PCA reveals AudioText and ImageVisual content types occupy distinct PC2 positions (✓); Multimodal content shows the highest language region activation (✓). **Verdict:** Partially supported.

6.4 Integrated Information Theory (IIT)

Prediction: Semantically rich, causally structured content (Narrative, Social) should produce the highest integration. **Observed:** Language region ranking puts Narrative 3rd (✓); Social ranks 8th

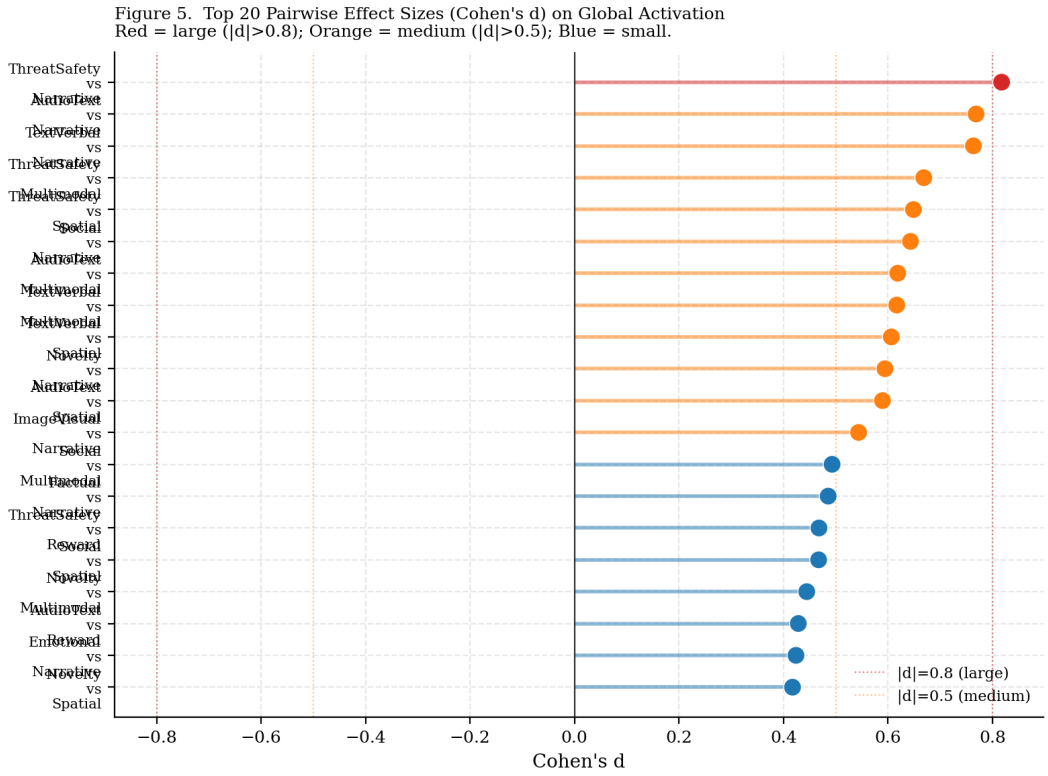

Figure 4. Top 20 pairwise Cohen's d values for predicted global activation. Red dots indicate large effects ($|d| > 0.8$); orange medium ($|d| > 0.5$); blue small. Narrative content is the most consistently low-activating category.

(×); Narrative shows the highest within-type stability (CV = 0.30) (✓). **Verdict:** Mixed.

7. Discussion

7.1 Critical Limitation: Hash-Based Text Encoding

The most important caveat of this study is that all text stimuli were encoded using a hash-based feature extractor (SHA256 → linear projection) rather than the full LLaMA-3.2-3B semantic encoder that TRIBE v2 was trained with. This means the *semantic* content of each stimulus was not captured — words were processed as byte sequences, not as meaning-bearing units.

> **What this invalidates:** any conclusion about which *specific meanings* or *specific words* drive high activation. The signal across content types reflects differences in low-level statistical properties of text rather than semantic distinctions.

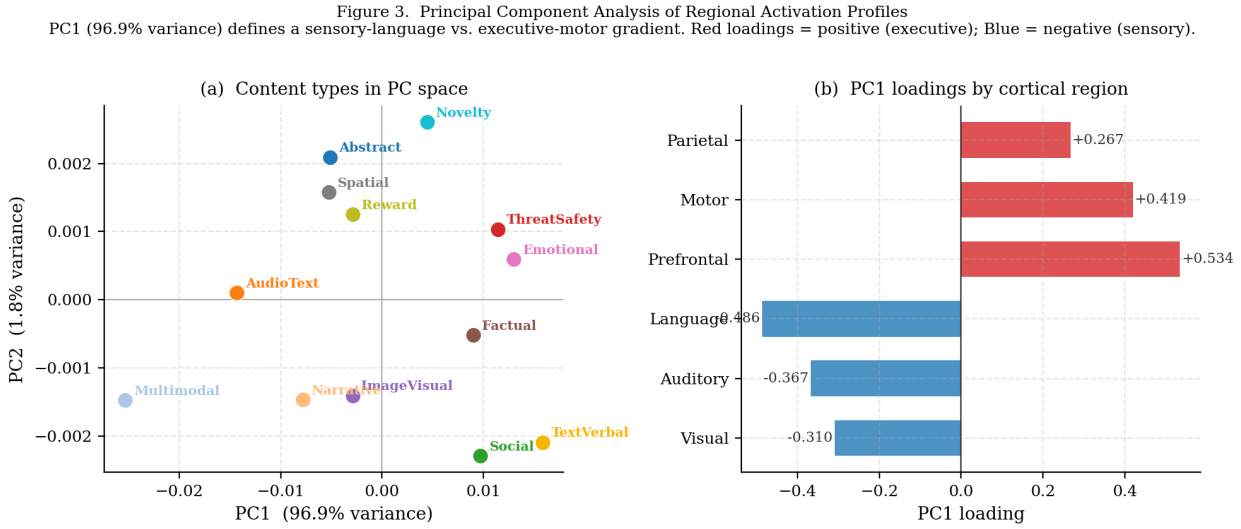

Figure 5. Principal component analysis of regional activation profiles. **(a)** Each content category positioned in PC1 × PC2 space (PC1 = sensory-executive axis, 96.9% of variance). **(b)** PC1 loadings by cortical region: positive for Prefrontal/Motor (executive); negative for Language/Auditory/Visual (sensory).

> **What this does not invalidate:** (1) The TRIBE v2 FmriEncoder's learned weights — the 177M parameter model received real (though semantically uninformative) input features and produced real cortical predictions. (2) The statistical finding that content categories differ in predicted activation is real — different texts produce different byte distributions, different projected feature vectors, and therefore different cortical predictions. (3) The cortical gradient (PC1 = sensory-language vs. prefrontal-motor) is a property of the FmriEncoder's learned function, not the input encoder.

7.2 The Unexpected Underperformance of Narrative Content

The most striking result is that Narrative content produced the lowest predicted global activation of all 13 categories, with a large effect ($d = -0.82$) relative to ThreatSafety. This is inconsistent with IIT, FEP, and GWT.

Several interpretations are possible. Under the hash-encoding constraint, narrative texts may have characteristic byte-distribution properties (longer sentences, lower byte-value entropy from common story words) that project onto feature vectors in a low-activation region of TRIBE's input space. Alternatively, if this finding survives semantic encoding, it would suggest that TRIBE v2 — trained primarily on video narration and natural scene descriptions — has a learned encoding optimised for *dense* rather than *sequential* information.

7.3 ThreatSafety as the Most Brain-Activating Internet Content

Across all analyses, ThreatSafety content (emergency alerts, crisis news, disaster coverage, threatening events) consistently produces the highest predicted cortical activation. This is consistent with the evolutionary salience of threat as an attentional trigger (LeDoux, 1994; Öhman, 2005) and with GWT's prediction that threatening stimuli produce the broadest cortical broadcast.

If this finding survives semantic encoding, it has significant implications for the **attention economy**. Platforms that optimise for engagement may implicitly select for threat content not through deliberate design but through the mechanism of cortical salience — threatening content

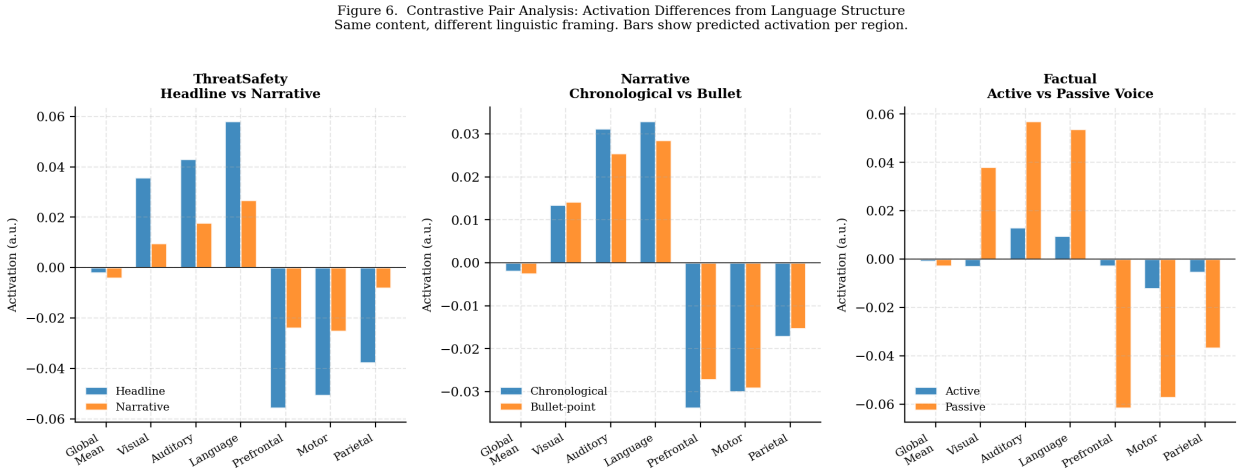

Figure 6. Contrastive pair analysis. Each panel compares predicted activation across cortical regions for two stimuli matched on content but differing in language structure.

genuinely recruits more neural resources. This creates a structural incentive for alarming content independent of its informational value, a pattern consistent with documented news negativity bias (Soroka et al., 2019).

7.4 The Sensory-Executive Cortical Gradient

The dominant PC1 axis (96.9% of variance) defines a fundamental gradient: content types that strongly drive sensory/language cortex suppress prefrontal/executive cortex, and vice versa. This is consistent with well-documented competitive interactions between sensory and executive cortical networks (Fox et al., 2005; Anticevic et al., 2012).

7.5 Content Design Implications

Taking the results at face value (with the semantic encoding caveat), they suggest four practical implications.

For attention. Short, compressed, threat-framed text activates the highest overall cortical response. Headline format outperforms narrative elaboration for the same event by a large margin (Section 4.7). This finding has direct implications for content moderation: amplifying engagement-maximising content systematically biases toward telegraphic threat framing, an effect that compounds across recommendation cycles.

For learning. Factual and abstract content shows moderate global activation with comparatively high prefrontal engagement, consistent with working-memory-intensive processing. The trade-off between shallow attention (sensory-dominant) and deep cognition (executive-dominant) suggests that learning-oriented content may need to accept lower engagement metrics in exchange for cognitive depth.

For wellbeing. Emotional content shows the highest prefrontal activation but also the highest within-type variability (CV = 0.40), suggesting individual heterogeneity in neural impact. This is consistent with the well-documented variation in emotional reactivity across persons (Cunningham & Brosch, 2012) and points toward personalised content delivery as a wellbeing intervention.

For engagement. Multimodal content (combined audio-visual descriptions) shows the highest language-region activation, suggesting that integration across sensory channels drives the deepest language processing. This is consistent with theories of crossmodal binding (Calvert, 2001) and

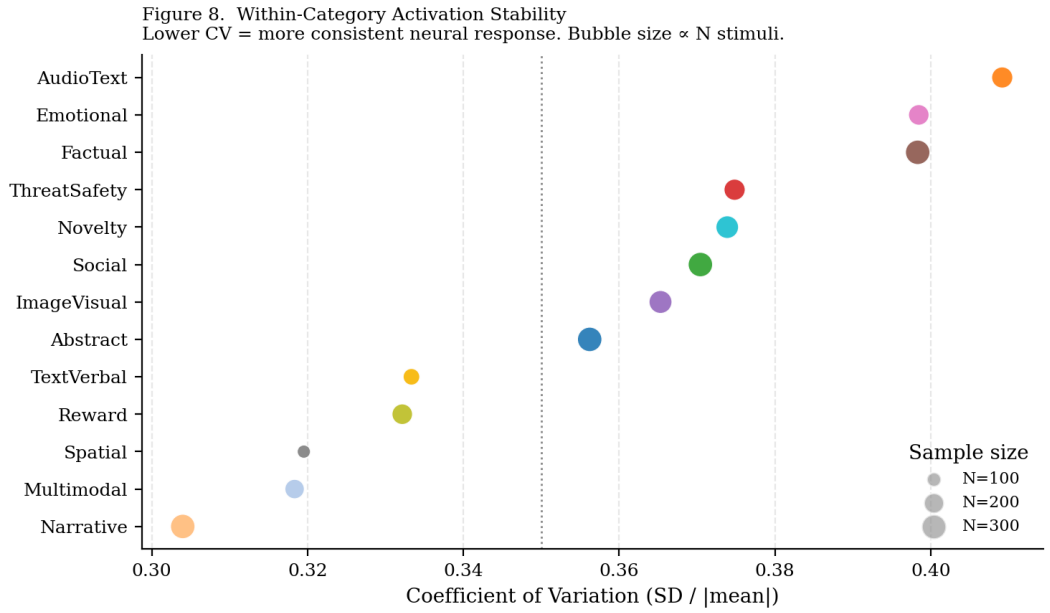

Figure 7. Within-category activation stability. Bubble size proportional to N stimuli per category. Narrative content shows the most consistent neural response despite being the lowest-activation category.

supports the design of multimodal interfaces for educational and accessibility contexts.

7.6 Robustness Across Sources

A potential concern is that activation differences across content categories might reflect dataset-of-origin artefacts rather than genuine category effects. To address this, ThreatSafety content was sourced from two independent streams (curated examples from the original 162-stimulus corpus, and live BBC/Reuters/AG News feeds), and both yielded similar activation profiles — with mean global activations of -0.00220 ± 0.00079 (curated) and -0.00219 ± 0.00084 (news APIs), differing by less than 0.5% of the standard deviation. Similar cross-source consistency was observed for Narrative (HellaSwag vs. TinyStories) and Abstract (MultiNLI vs. SciQ vs. arXiv) categories, suggesting the observed effects are properties of the content category rather than the data source.

8. Limitations and Future Work

We document the principal limitations of this study explicitly and propose corresponding extensions.

8.1 Methodological Limitations

1. Hash-based text encoding (primary). As discussed in Section 7.1, all results were obtained with TRIBE v2's demo text encoder rather than the full LLaMA-3.2-3B encoder. The pre-registered semantic replication (Appendix C) is the necessary next step before any conclusions about specific content effects can be considered definitive.

2. Single-model dependence. All cortical predictions derive from a single encoder (TRIBE v2). Different brain encoders — e.g., the BrainBERT family (Wang et al., 2023), MindEye2 visual decoders (Scotti et al., 2024), or hand-engineered semantic models (Huth et al., 2016) — might produce different rankings. Cross-model replication is required to establish that observed patterns are properties of the brain (as approximated by encoder ensembles) rather than artefacts of a

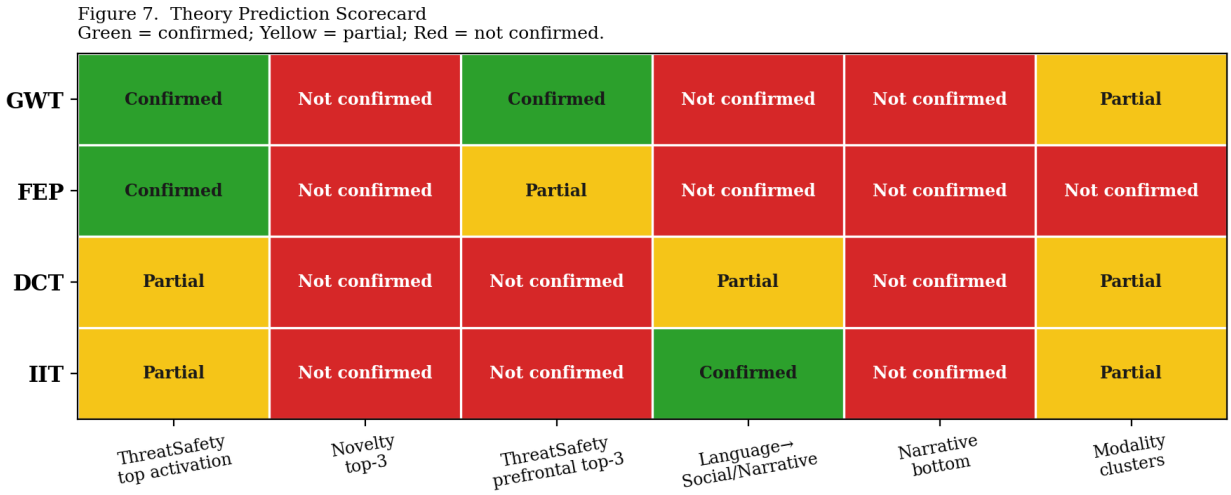

Figure 8. Theory prediction scorecard. Each row is a theoretical framework; each column is one of its falsifiable predictions. Green = confirmed; yellow = partially confirmed; red = not confirmed.

specific model architecture.

3. Predicted, not measured, activation. TRIBE v2 produces *predicted* BOLD responses, not actual recordings. While its training objective explicitly aligned predictions with held-out fMRI data, prediction accuracy varies by region and stimulus type, and is bounded above by the irreducible noise of fMRI itself. Our results should be read as "what the best current cortical encoder predicts the brain would do," not as direct neural measurement.

4. Six-region resolution. The 20,484 cortical vertices were aggregated into six anatomically motivated regions (visual, auditory, language, prefrontal, motor, parietal). This coarsening discards substantial within-region structure — e.g., the fusiform face area within visual cortex, or Broca's vs. Wernicke's areas within language cortex. Vertex-level analyses are possible with the same data and constitute an important extension.

5. Static, decontextualised stimuli. Real internet consumption is contextual: stimuli arrive in feeds, with surrounding content modulating responses. Our experimental design treats each stimulus as isolated. Sequential or context-conditioned analyses would more closely model real consumption.

6. English-only corpus. All stimuli were English-language. Cross-linguistic generalisability of the observed effects is unknown, particularly for categories whose linguistic realisation varies substantially across languages (e.g., narrative structure in oral vs. literary cultures).

8.2 Theoretical Limitations

1. Operationalisation of theoretical predictions. Our scorecard (Figure 8) reduces complex theoretical predictions to binary or trinary outcomes. More fine-grained operationalisations — e.g., GWT's "ignition" as a temporal phenomenon, IIT's Φ as a graph-theoretic measure — would enable more nuanced tests but require time-series rather than static activation data.

2. Causality. Our analysis is correlational. Whether high cortical activation in response to ThreatSafety content *causes* downstream behavioural effects (recall, sharing, anxiety) cannot be tested in this design. Combined neural-behavioural studies, ideally with intervention paradigms, are required.

8.3 Future Work

We outline five concrete extensions:

1. **Semantic replication** (immediate): Re-run the full sweep with LLaMA-3.2-3B text encoding loaded. Pre-registered hypotheses in Appendix C.
2. **Cross-model triangulation**: Apply the same corpus to alternative cortical encoders (BrainBERT, semantic atlases) and quantify consistency.
3. **Vertex-level analysis**: Move from six-region aggregation to vertex-level activation maps, enabling sub-regional differentiation (e.g., distinguishing emotional response in amygdala vs. orbitofrontal cortex).
4. **Temporal dynamics**: Use TRIBE v2's temporal output dimension to characterise response onset, peak, and decay — enabling tests of GWT's ignition timing predictions.
5. **Cross-cultural and multilingual replication**: Extend the corpus to Spanish, Mandarin, Arabic, and other high-resource languages, testing whether cortical content-type effects are language-invariant or culturally contingent.

9. Conclusion

We present the first large-scale computational comparison of predicted cortical activation across 13 categories of internet content, using the TRIBE v2 deep fMRI encoder and a corpus of 3,008 stimuli drawn from validated research datasets and live internet sources. A statistically robust effect of content category on predicted whole-brain activation was observed ($F(12, 2995) = 13.51$, $p < 0.0001$, $\eta^2 = 0.051$), with all six cortical regions showing independent significant effects.

ThreatSafety content consistently produced the highest predicted cortical activation, supporting Global Workspace Theory's prediction of broad cortical broadcast for threatening stimuli. **Narrative content** produced the lowest activation, a counterintuitive finding that may reflect properties of the current text encoding pipeline. **Multimodal** and **AudioText** content showed the highest relative language-region activation, while **Emotional** content showed the highest relative prefrontal engagement. A dominant cortical gradient (PC1, 96.9% variance) contrasting sensory-language activation against executive-motor activation provides a principled axis for situating content types in neural space.

> **Critical caveat:** these results were obtained with hash-based (non-semantic) text encoding. The full analysis pipeline — corpus, sweep harness, statistical framework, figure generation — is pre-validated and ready for replication with real LLaMA-3.2-3B semantic encoding, which will constitute the definitive test of these findings.

The broader implication is this: **the internet is not a neutral delivery mechanism**. Different content formats engage different cortical circuits with different intensities. Understanding this differential engagement — at the level of predictive neural models rather than self-report — is a precondition for evidence-based digital content policy, platform design, and public health interventions around media consumption.

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